

Exercise (Full reconstruction from bits). Let $\mathcal{M}$ be of a the messages space of size 2^n and let predicates $\psi_1, \dots, \psi_n : \mathcal{M} \rightarrow \{0, 1\}$ be such that their outcome uniquely determines the message space element, i.e, the system of equations

$$\psi_1(m) = b_1, \dots, \psi_n(m) = b_n$$

has a unique solution. Assume that for any t -time adversary the probability that $\mathcal{A}$ wins a game

$$\mathcal{G}_i \left[\begin{array}{l} \text{sk} \leftarrow \text{Gen} \\ m \leftarrow \mathcal{M} \\ c \leftarrow \text{Enc}_{\text{sk}}(m) \\ \text{return } [\psi_i(m) \stackrel{?}{=} \mathcal{A}(c)] \end{array} \right.$$

is at least $\frac{1}{2} + \varepsilon$ for any $i \in \{1, \dots, n\}$. How large ε has to be so that one can prove that there exists an efficient algorithm $\mathcal{B}$ that can reconstruct m from an encryption, i.e., win the following game

$$\mathcal{Q} \left[\begin{array}{l} \text{sk} \leftarrow \text{Gen} \\ m \leftarrow \mathcal{M} \\ c \leftarrow \text{Enc}_{\text{sk}}(m) \\ \text{return } [m \stackrel{?}{=} \mathcal{B}(c)] \end{array} \right.$$

with non-trivial probability provided that 2^n is small enough to be enumerated and $\psi_1, \dots, \psi_n$ are efficiently computable. Is it possible to define $\psi_1, \dots, \psi_n$ so that the claim holds even $t \ll 2^n$? What additional assumption allows us to prove the claim for any $\varepsilon > 0$?

Solution.

Hint: What is trivial success for the game $\mathcal{Q}$?

Hint: You can fix a sensible construction of $\mathcal{B}$.

Hint: Define $\varepsilon_i(m)$ as the probability that $\mathcal{A}(m)$ guesses $\psi_i(m)$. Is it possible $\varepsilon_1(m) \cdots \varepsilon_n(m) = 0$? What does it mean for the success of $\mathcal{B}$? Analyze this for $n = 2$, $n = 3$ and generalise.